

Future Shellf

Predictive Phenotyping for Oyster Performance

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SORMI.SCIENCE

Two questions

- Which seed should I buy — which family lines actually perform on my site?
- How stressed is my crop right now, before I start seeing mortality?

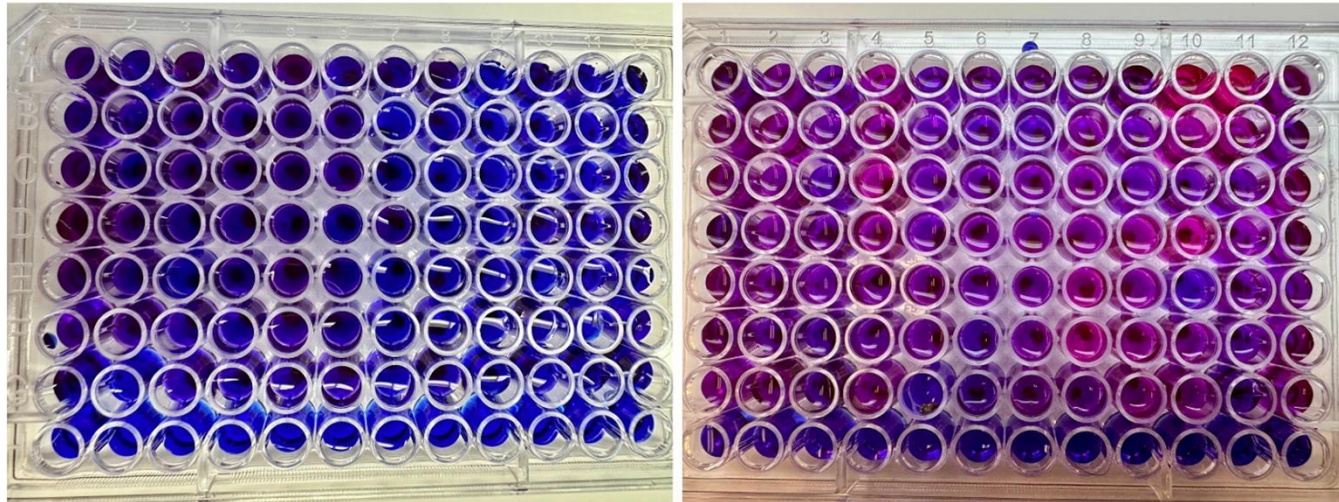
Today, answering either one means waiting.

Breeding trials take a season. Mortality events tell you after the fact.

We need a fast, cheap, non-lethal measure of how an oyster is doing — right now.

From blue to pink: the resazurin assay

A redox dye that reports whole-organism metabolism



Resazurin is blue and non-fluorescent.

Living cells reduce it to resorufin — pink and strongly fluorescent.

More metabolism = more pink = more signal.

The animal stays alive. Nothing is sacrificed.

From blue to pink: the resazurin assay

Huffmyer et al. 2026, PeerJ 14:e21542 — DOI 10.7717/peerj.21542

- Resazurin fluorescence correlates with oxygen consumption on the same individuals.
- Signal comes from live animals — live oysters far exceed empty shells and blanks.
- Scales with body size, so we normalize to shell length.
- Thermal performance curves behave as expected: metabolism peaks near 36 °C, then depresses.
- Works in both Pacific (*C. gigas*) and Eastern (*C. virginica*) oysters, seed through juveniles.

Peer-reviewed, open access, protocol included as an appendix — anyone can run it.

Use #1 — Predicting phenotype

Read metabolism today, forecast field performance later

Metabolism already tracks outcomes

Key findings from the PeerJ study

- Under acute heat stress, oysters that survived had ~52% lower cumulative metabolism than those that died.
- High metabolic output under stress = higher risk of mortality ($p < 0.001$).
- Families differ significantly in metabolic response — the trait is genetically variable.
- In 50 selectively bred *C. virginica* families, metabolism correlated with predicted survival in the environment they were selected for ($r = 0.56$, $p = 0.004$).

Metabolism is not just a health readout — it carries information about future performance.

Testing the idea head-on: survival trials

Nine USDA C. gigas families, run side by side

Trial	Stressor	Oysters	% died
Apr 27	33 °C	128	49%
May 11	33 °C	84	99%
Jun 04	36 °C	39	95%
Jun 09	36 °C	77	86%
Jul 06	35 °C	378	97%
Jul 06	35 °C + low salinity	204	84%
Jul 07	Freshwater → 36 °C	185	33%

Harshness varies enormously — raw survival can't be compared across trials.

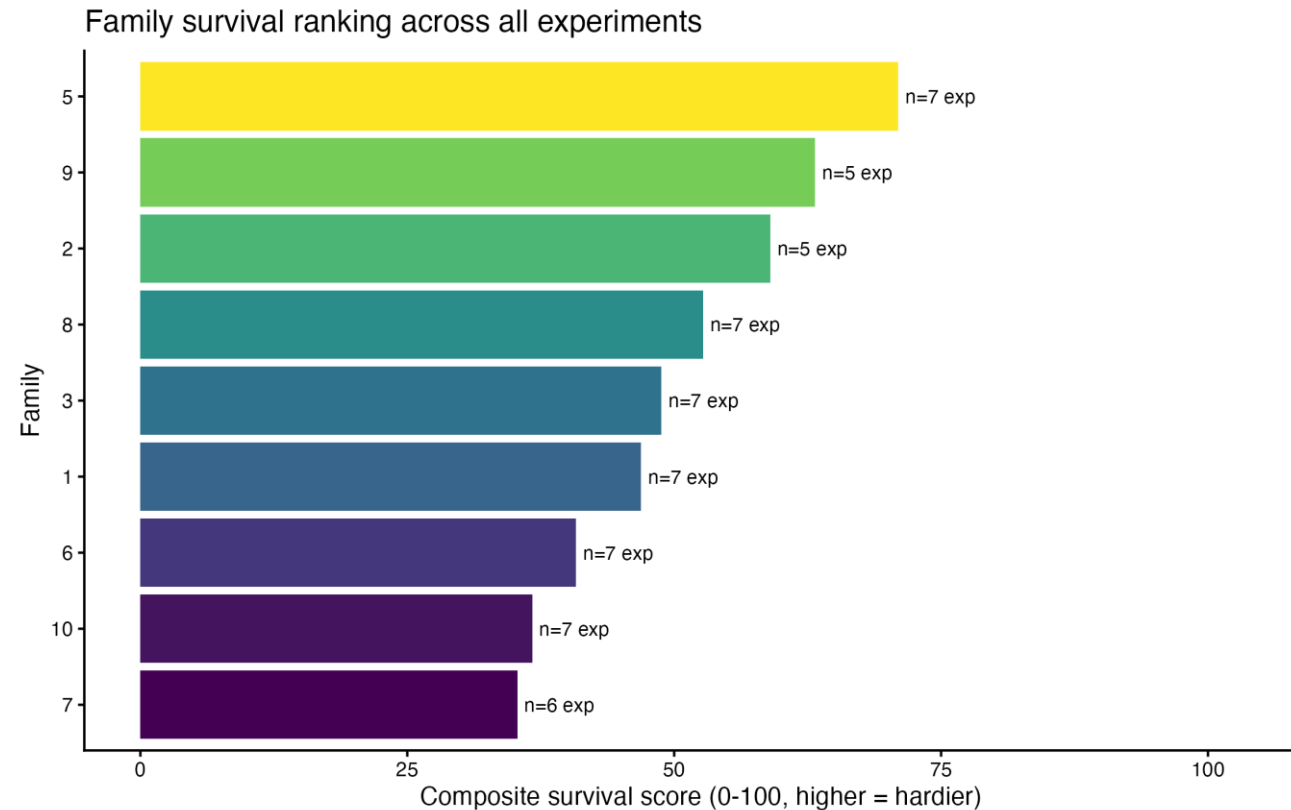
A fair hardiness score

0–100, comparable across trials

- Within each trial, rank families on (1) fraction surviving and (2) how long they lasted.
- Convert each family's standing to a percentile against the other families in that same trial — this cancels out trial harshness.
- Average those percentiles across all seven trials.

Higher score = consistently tougher across many kinds of stress.

Families really do differ



Family 5 is the toughest; family 7 the most fragile — and the spread repeats across trials.

Predictive Phenotyping

We have two independent results on the **same 9 families**:

- **A survival score** (from the lethal trials).
- **A metabolism fingerprint** (from the non-lethal dye test).

Question: can the metabolism fingerprint predict the survival score — provide data quicker?

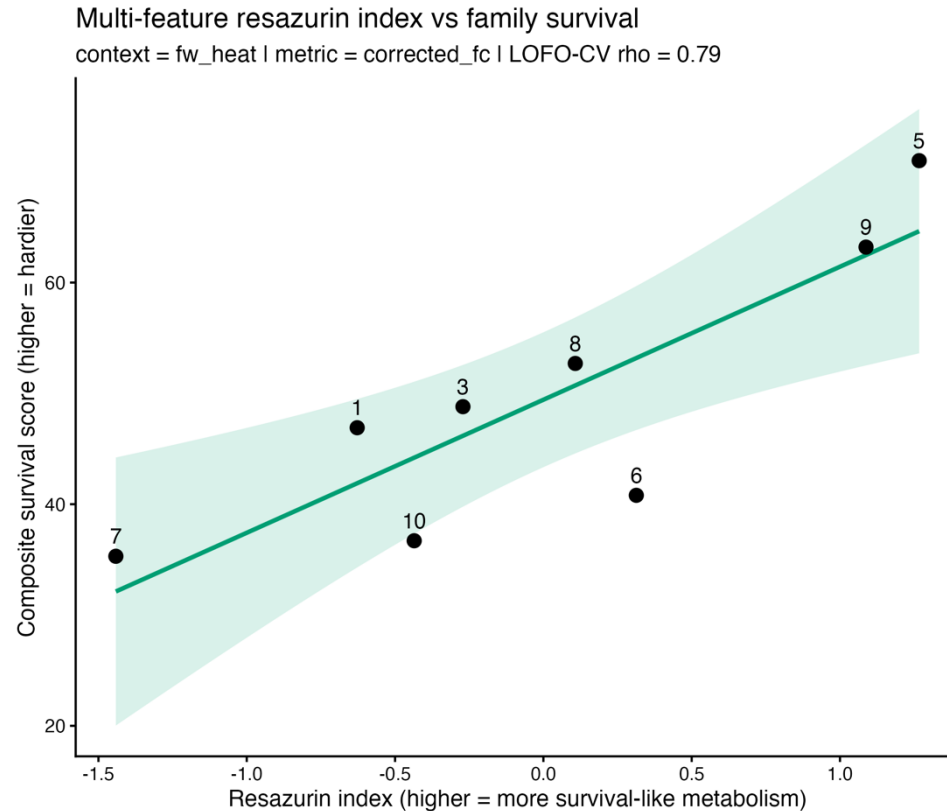
The resazurin index

Four curve features, blended into one score per family

Feature	Plain meaning	Survivor-like value
Time to steepest drop	how long before metabolism crashes	later
Steadiness	how much metabolism bounces around	steadier
Early rise	how fast metabolism climbs at first	faster
Depth of collapse	how far it falls from its peak	smaller

Each feature is put on a common scale, flipped so higher = tougher, and the four are averaged.

The index tracks real survival



- X: combined resazurin index (further right = more survivor-like metabolism).
- Y: survival score from the lethal trials.
- Leave-one-family-out rho = 0.79 — strong prediction for families the model never saw.
- Toughest (5) and most fragile (7) land at opposite ends.

Use #2 — Real-time stress on the farm

A thermometer for physiological condition - early development – conceptual

Reading stress while it's happening

Same assay, different question

- Pull a small sample of animals, run a plate, get an answer the same day.
- Elevated metabolism under a heat or salinity event flags a crop under load — before mortality shows up.
- Compare bags, tidal heights, gear types, or sites on the same afternoon.
- Track recovery after an event: does metabolism return to baseline, and how fast?
- Non-lethal and cheap enough to repeat weekly through a risky season.

Turns "the water got hot last week" into a number you can act on.

Takeaways

- Resazurin turns metabolism into a color change — cheap, non-lethal, 96 animals at a time.
- Predicts phenotype: metabolic curve shape forecasts family hardiness.
- Need additional field performance data and will add other assays to index.
- Could measure stress in real time: a same-day readout of how hard your crop is working.

USDA-WRAC (Hudson - PSI)

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Stress Resilience Assay Panel

Five candidate traits were selected based on established stress-response literature, low cost (<\$10/sample), and compatibility with high-throughput 96-well plate formats suitable for breeding programs and on-farm evaluation.

Resazurin Metabolic Rate

Organismal metabolic rate

Non-lethal, colorimetric measure of whole-organism metabolism. Lower metabolic rate under stress and faster recovery predict field performance.

< \$1.00 / sample

Na⁺/K⁺ ATPase

Cellular stress management

Transmembrane pump activity essential for osmotic regulation. Higher activity is positively associated with stress resilience.

~\$2.50 / sample

Glycogen Content

Energy storage

Critical carbohydrate stores fueling metabolism and reproduction. Higher baseline glycogen and retention under stress correlate with survival.

~\$5.00 / sample

Citrate Synthase

Aerobic metabolic capacity

Marker of oxidative metabolic potential. Higher activity indicates greater capacity to generate energy under stress.

~\$5.70 / sample

Lactate Dehydrogenase

Anaerobic metabolic capacity

Measures anaerobic energy production when oxygen is limited. Higher LDH activity is expected to improve stress tolerance.

~\$4.10 / sample

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